



Shaheed Zulfikar Ali Bhutto Institute of Science & Technology

COMPUTER SCIENCE DEPARTMENT

Total Marks: _____

Marks Obtained: _____

Applied Data Mining

Assignment # 04

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Objective of the Assignment

Instructions

- The assignment must be handwritten/computer written.
- Show complete steps for all calculations.
- Late submissions will not be accepted.
- If two assignments are found identical, both will be awarded zero marks.
- Any assignment found copied from AI tools such as ChatGPT, Gemini, Copilot, or other similar tools will be marked zero.
- File name: (YourRollNumber.pdf)

Question 1

Consider the following 2D data points:

Point Coordinates	
A1	2, 10
A2	2, 5
A3	8, 4
A4	5, 8
A5	7, 5
A6	6, 4
A7	1, 2
A8	4, 9

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Using **K-Means Clustering** with:

- Number of clusters $k = 3$
- Initial centroids:

- $C_1 = (2,10)$
- $C_2 = (5,8)$
- $C_3 = (1,2)$

Perform **one complete iteration of K-Means** using the Manhattan distance formula:

$$d(a, b) = |x_2 - x_1| + |y_2 - y_1|$$

You must:

1. Calculate the distance of each point from all centroids.
2. Assign each point to the nearest cluster.
3. Compute the new centroids after clustering.

Solution

We use K-Means Clustering with:

- $k = 3$
- Initial Centroids:
 - $C_1 = (2,10)$
 - $C_2 = (5,8)$
 - $C_3 = (1,2)$

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Manhattan Distance Formula:

$$d(a, b) = |x_2 - x_1| + |y_2 - y_1|$$

Step 1: Calculate Distance from Each Centroid

Point A1 = (2,10)

- From C_1 :

$$|2 - 2| + |10 - 10| = 0$$

- From C_2 :

$$|5 - 2| + |8 - 10| = 3 + 2 = 5$$

- From C_3 :

$$|1 - 2| + |2 - 10| = 1 + 8 = 9$$

Nearest Cluster $\rightarrow$ **C1**

Point A2 = (2,5)

- C_1 : $|2 - 2| + |10 - 5| = 0 + 5 = 5$
- C_2 : $|5 - 2| + |8 - 5| = 3 + 3 = 6$
- C_3 : $|1 - 2| + |2 - 5| = 1 + 3 = 4$

Nearest Cluster $\rightarrow$ **C3**

Point A3 = (8,4)

- C_1 : $|2 - 8| + |10 - 4| = 6 + 6 = 12$
- C_2 : $|5 - 8| + |8 - 4| = 3 + 4 = 7$
- C_3 : $|1 - 8| + |2 - 4| = 7 + 2 = 9$

Nearest Cluster $\rightarrow$ **C2**

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Point A4 = (5,8)

- $C_1: |2 - 5| + |10 - 8| = 3 + 2 = 5$
- $C_2: |5 - 5| + |8 - 8| = 0$
- $C_3: |1 - 5| + |2 - 8| = 4 + 6 = 10$

Nearest Cluster → **C2**

Point A5 = (7,5)

- $C_1: |2 - 7| + |10 - 5| = 5 + 5 = 10$
- $C_2: |5 - 7| + |8 - 5| = 2 + 3 = 5$
- $C_3: |1 - 7| + |2 - 5| = 6 + 3 = 9$

Nearest Cluster → **C2**

Point A6 = (6,4)

- $C_1: |2 - 6| + |10 - 4| = 4 + 6 = 10$
- $C_2: |5 - 6| + |8 - 4| = 1 + 4 = 5$
- $C_3: |1 - 6| + |2 - 4| = 5 + 2 = 7$

Nearest Cluster → **C2**

Point A7 = (1,2)

- $C_1: |2 - 1| + |10 - 2| = 1 + 8 = 9$
- $C_2: |5 - 1| + |8 - 2| = 4 + 6 = 10$
- $C_3: |1 - 1| + |2 - 2| = 0$

Nearest Cluster → **C3**

Point A8 = (4,9)

- $C_1: |2 - 4| + |10 - 9| = 2 + 1 = 3$
- $C_2: |5 - 4| + |8 - 9| = 1 + 1 = 2$
- $C_3: |1 - 4| + |2 - 9| = 3 + 7 = 10$

Nearest Cluster → **C2**

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Step 2: Cluster Assignment

Cluster	Points
C1	A1
C2	A3, A4, A5, A6, A8
C3	A2, A7

Step 3: Compute New Centroids

Centroid Formula:

$$\left(\frac{\sum x}{n}, \frac{\sum y}{n} \right)$$

New Centroid for C1

Points: A1(2,10)

$$C'_1 = (2,10)$$

New Centroid for C2

Points:

- A3(8,4)
- A4(5,8)
- A5(7,5)
- A6(6,4)
- A8(4,9)

Mean X-coordinate

$$\frac{8 + 5 + 7 + 6 + 4}{5} = \frac{30}{5} = 6$$

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Mean Y-coordinate

$$\frac{4 + 8 + 5 + 4 + 9}{5} = \frac{30}{5} = 6$$

$$C'_2 = (6,6)$$

New Centroid for C3

Points:

- A2(2,5)
- A7(1,2)

Mean X-coordinate

$$\frac{2 + 1}{2} = 1.5$$

Mean Y-coordinate

$$\frac{5 + 2}{2} = 3.5$$

$$C'_3 = (1.5,3.5)$$

Final Answer

Cluster Assignments

Cluster	Assigned Points
C1	A1
C2	A3, A4, A5, A6, A8
C3	A2, A7

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New Centroids After One Iteration

Cluster	New Centroid
C1	(2,10)
C2	(6,6)
C3	(1.5,3.5)

Question 2

Consider the following distance matrix:

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

Apply **Single Linkage Agglomerative Clustering** and:

1. Show all cluster merging steps.
2. Compute updated distance matrices after each merge.
3. Draw the final dendrogram.

Single Linkage Formula:

$$d(A, B) = \min \{d(x, y): x \in A, y \in B\}$$

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Step 1: Initial Distance Matrix

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

Step 2: First Merge

Smallest distance:

$$d(c, e) = 2$$

Merge:

$$(c, e)$$

New cluster:

$$CE = \{c, e\}$$

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Updated Distance Matrix After Merge (CE)

Using single linkage:

$$d(CE, a) = \min(d(c, a), d(e, a))$$

$$= \min(3, 11) = 3$$

$$d(CE, b) = \min(7, 10) = 7$$

$$d(CE, d) = \min(9, 8) = 8$$

Updated matrix:

	a	b	d	CE
a	0	9	6	3
b	9	0	5	7
d	6	5	0	8
CE	3	7	8	0

Step 3: Second Merge

Smallest distance:

$$d(a, CE) = 3$$

Merge:

$$(a, CE)$$

New cluster:

$$ACE = \{a, c, e\}$$

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Updated Distance Matrix After Merge (ACE)

Smallest distance:

$$d(a, CE) = 3$$

Merge:

$$(a, CE)$$

New cluster:

$$ACE = \{a, c, e\}$$

	b	d	ACE
b	0	5	7
d	5	0	6
ACE	7	6	0

Step 4: Third Merge

Smallest distance:

$$d(b, d) = 5$$

Merge:

$$(b, d)$$

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New cluster:

$$BD = \{b, d\}$$

Updated Distance Matrix After Merge (BD)

Compute distance:

$$d(ACE, BD) = \min(7, 6) = 6$$

Updated matrix:

	ACE	BD
ACE	0	6
BD	6	0

Step 5: Final Merge

Smallest distance:

$$d(ACE, BD) = 6$$

Merge all clusters.

Final cluster:

$$\{a, b, c, d, e\}$$

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Complete Cluster Merging Steps

Step	Clusters Merged	Distance
1	(c, e)	2
2	(a, CE)	3
3	(b, d)	5
4	(ACE, BD)	6

Question 3

Answer the following conceptual questions briefly:

(a) Differentiate between:

- Cluster Analysis
- Clustering

(b) Explain the following terms with examples:

- High intra-cluster similarity
- Low inter-cluster similarity

(c) Compare:

- Partitioning Clustering
- Hierarchical Clustering

(Write at least 3 differences)

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Difference Between Cluster Analysis and Clustering

Cluster Analysis	Clustering
Cluster analysis is the overall process of studying and grouping data.	Clustering is the technique used to create groups (clusters).
It includes evaluation, interpretation, and analysis of clusters.	It mainly focuses on dividing data into clusters.
It is a broader concept in data mining and machine learning.	It is one step within cluster analysis.

Explanation of Terms

1. High Intra-Cluster Similarity

High intra-cluster similarity means the data points inside the same cluster are very similar to each other.

Example:

Students with similar marks grouped into one cluster.

For example:

[(85,90), (88,92), (84,89)]

These points are close to each other, so they have high similarity.

2. Low Inter-Cluster Similarity

Low inter-cluster similarity means different clusters are very different from each other.

Example:

One cluster contains students scoring above 85, while another contains students scoring below 40.

The clusters are far apart and not similar.

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Difference Between Partitioning Clustering and Hierarchical Clustering

Partitioning Clustering	Hierarchical Clustering
Divides data into a fixed number of clusters (k).	Creates a hierarchy/tree of clusters.
Example: K-Means algorithm.	Example: Agglomerative clustering.
Works faster for large datasets.	Slower for large datasets.
Requires the value of (k) before execution.	Does not always require predefined (k).
Produces one-level clustering.	Produces multi-level clustering using a dendrogram.
Clusters can change during iterations.	Once merged/split, clusters usually cannot be changed.